

DTSA 5508

Software Architecture Patterns for Big Data

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	MLOps Engineer (100.0 / 100)	91.8 / 100
SPECIALIZATION Software Development Lifecycle for Data Scientists		

92

CURRENT AVG SCORE
across 6 job roles

100

POTENTIAL SCORE
with recommended additions

+25

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Intended for individuals looking to understand the architecture patterns necessary to take large software systems that leverage big data to production. Students will transform big data prototypes into high quality tested production software. After measuring the performance characteristics of distributed systems, they will identify trouble areas and implement scalable solutions to improve performance. Upon completion of the course they will know how to scale production datastores to perform under load, designing load tests to ensure applications meet performance requirements. Formerly offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
MLOps Engineer	100.0	100.0	+0.0	LOW
Machine Learning Engineer	100.0	100.0	+0.0	LOW
AI Engineer	99.4	100.0	+0.6	LOW
ML Engineer	97.0	100.0	+3.0	LOW
Data Scientist	78.8	100.0	+21.2	MEDIUM
Applied Scientist	75.3	100.0	+24.7	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication 2 of 6 roles

insightsbusiness

LLMs & Generative AI 2 of 6 roles

storage

Feature Engineering 1 of 6 roles

preprocessing

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

MEDIUM

Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.
- 2

MEDIUM

Consider LLMs & Generative AI content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.
- 3

LOW

Optional: Feature Engineering — would broaden appeal to one additional role. Candidate terms: preprocessing.

GAP KEYWORDS BY ROLE

Data Scientist

78.8 → 100.0 (+21.2 pts)

setsproductanalytics evolutionscripting insights analyticalsocial

Applied Scientist

75.3 → 100.0 (+24.7 pts)

labdiscovery scientific researchstorage problemsmodels learning

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.